

EXCAVATION AND TRENCHING COMPETENT PERSON

General Safety

Each jobsite that has a trench or excavation shall have onsite an employee that has been trained as a Competent Person Training based on WAC 296-155 Part N or 1926 Subpart P.

The Competent Person shall supervise the safety of all excavation and trenching work and complete a daily trench inspection form for work being done.

Subcontractors performing work in trenches must provide evidence of Competent Person training for their Supervisors and daily Trench Inspection forms.

Training

All persons who work in trenches must be trained on all aspects of the **Trench Work Plan and Inspection Form** before starting work.

Trench Definition: A narrow excavation in relation to its length made below the surface of the ground. In general, the depth is greater than the width, but the width of a trench (measured at the bottom) is not greater than 15 feet (4.6m). If forms or other structures are installed or constructed in an excavation so as to reduce the dimension measured from the forms or structure to the side of the excavation to 15 feet (4.6 m) or less (measured at the bottom of the excavation), the excavation is also considered to be a trench.

Excavation Definition: Any person-made cut, cavity, trench, or depression in the earth's surface, formed by earth removal.

Competent Person

A Competent Person is one who can:

1. identify existing or predictable hazards in the surroundings that are unsanitary, hazardous, or dangerous to employees.
2. Also has authorization or authority by the nature of their position to take prompt corrective measures to eliminate them.
3. The person must be knowledgeable in the requirements of this part. WAC 296-155-PART N or 29 CFR 1926 Subpart P.

A Competent Person must determine soil type to design protective systems.

Soil Determinations

Type A – Cohesive soils with an unconfined compressive strength of 1.5 tons per square foot (tsf) (144 kPa) or greater. Examples include: clay, silty clay, sandy clay, and clay loam. Certain conditions preclude soil from being classified as Type A. For example, no soil is Type A if it is fissured or has been previously disturbed.

Type B – Includes cohesive soil with an unconfined compressive strength greater than 0.5 tsf (48 kPa) but less than 1.5 tsf (144 kPa) and granular cohesionless soils (such as angular gravel, similar to crushed rock, silt, silt loam, sandy loam, and, in some cases, silty clay loam and sandy clay loam).

Type C – Cohesive soil with an unconfined compressive strength of 0.5 tsf (48 kPa) or less, granular soils (including gravel, sand, and loamy sand), submerged soil or soil from which water is freely seeping, submerged rock that is not stable, or material in a sloped, layered system where the layers dip into the excavation or with a slope of four horizontal to one vertical (4H:1V) or steeper.

Stable rock - A natural solid mineral material that can be excavated with vertical sides and will remain intact while exposed. Unstable rock is considered to be stable when the rock material on the side or sides of the excavation is secured against caving-in or movement by rock bolts or by another protective system that has been designed by a registered professional engineer.

If you can't accurately determine the soil type you must assume its type C.

Use one of the following tests to determine soil type:

(a) Visual tests. Visual analysis is conducted to determine qualitative information regarding the excavation site in general, the soil adjacent to the excavation, the soil forming the sides of the open excavation, and the soil taken as samples from excavated material.

EXCAVATION AND TRENCHING COMPETENT PERSON

(b) Manual tests. Manual analysis of soil samples is conducted to determine quantitative as well as qualitative properties of soil and to provide more information in order to classify soil properly.

(i) Plasticity. Mold a moist or wet sample of soil into a ball and attempt to roll it into threads as thin as 1/8-inch in diameter. Cohesive material can be successfully rolled into threads without crumbling. For example, if at least a two inch (50 mm) length of 1/8-inch thread can be held on one end without tearing, the soil is cohesive.

(ii) Dry strength. If the soil is dry and crumbles on its own or with moderate pressure into individual grains or fine powder, it is granular (any combination of gravel, sand, or silt). If the soil is dry and falls into clumps which break up into smaller clumps, but the smaller clumps can only be broken up with difficulty, it may be clay in any combination with gravel, sand or silt. If the dry soil breaks into clumps which do not break up into small clumps and which can only be broken with difficulty, and there is no visual indication the soil is fissured, the soil may be considered unfissured.

(iii) Thumb penetration. The thumb penetration test can be used to estimate the unconfined compressive strength of cohesive soils.) Type A soils with an unconfined compressive strength of 1.5 tsf can be readily indented by the thumb; however, they can be and penetrated by the thumb only with very great effort. Type C soils with an unconfined compressive strength of 0.5 tsf can be easily penetrated several inches by the thumb, and can be molded by light finger pressure. This test should be conducted on an undisturbed soil sample, such as a large clump of spoil, as soon as practicable after excavation to keep to a minimum the effects of exposure to drying influences. If the excavation is later exposed to wetting influences (rain, flooding), the classification of the soil must be changed accordingly.

(iv) Other strength tests. Estimates of unconfined compressive strength of soils can also be obtained by use of a pocket penetrometer or by using a hand-operated shear vane.

(v) Drying test. The basic purpose of the drying test is to differentiate between cohesive material with fissures, unfissured cohesive material, and

granular material. The procedure for the drying test involves drying a sample of soil that is approximately one inch thick (2.54 cm) and 6 inches (15.24 cm) in diameter until it is thoroughly dry:

Safety Reminders

A Trenching Work Plan and Inspection Form must be completed before allowing workers to enter a trench.

Always check with utility companies and call 811 before digging. Locate all underground utilities near the trenching operation.



Support any utilities adjacent to or crossing the trench. Overhead power lines are also a potential hazard.

Trench collapse kills workers. Keep unauthorized workers away from any open trench.

Use **temporary fencing** to keep workers who are not actively working in a trench away. The affected area must be at least as deep as the trench. (If the trench is up to 15' deep, the affected area is the depth of the trench to each side.

Walkways or bridges with standard railings must be provided when employees or equipment are required to cross over excavations.

The walls and faces of all excavations in which employees are exposed to danger from moving ground **must be guarded** by a shoring system, sloping of the ground, or some other equivalent means.

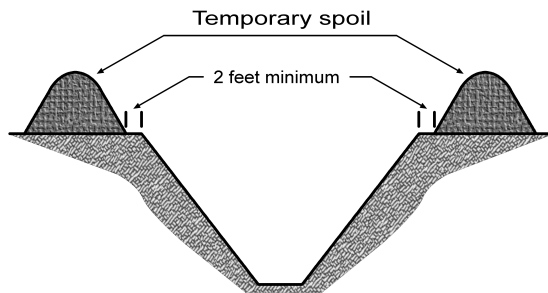
No person must be allowed under loads handled by excavators, backhoes, forklifts, derricks, or hoists. Swing radius should be marked and no person shall be allowed to work

EXCAVATION AND TRENCHING COMPETENT PERSON

or walk through that area without making eye contact with the operator.

Don't allow water to accumulate in a trench. Water reduces soil stability; its presence may cause you to lower the soil classification from type A to B or C, or type B to C.

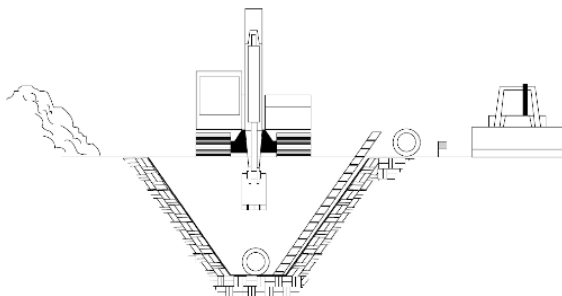
Keep **excavated materials at least 2' away** from the edge of the trench.



In trenches, deeper than 4', locate a **means of exit**, such as ladders or steps, so they are **no more than 25' of travel** from anywhere in the trench. When a box is used, a ladder for egress must be in the box at all times regardless of situation.

An earth ramp sloped at one end of the trench is considered a safe means of egress as long as employees are able to walk normally in an upright manner without assistance when entering or exiting the trench or excavation.

Vibrations from construction equipment, nearby construction operations, or traffic, etc., can create hazards. You may need to slope your trench less steeply, inspect your shoring more



often, and lower the soil classification from Type A to B or C, or Type B to C.

Store all materials at least 2' from the edge of the trench.

Keep heavy loads of all kinds as far from a trench as possible.

Air monitoring or ventilation may be required at a 4' depth if there is a reasonable potential for oxygen deficiency or a toxic atmosphere.

A Competent Person must inspect the trench, adjacent areas, and any protective systems for possible cave-ins, failure of protective systems, hazardous atmospheres, or other hazardous conditions.

Trench Inspections must be performed daily: before work begins, throughout the shift, and after every rainstorm or another hazard-increasing occurrence.

Hard hats must be worn in trenches at all times or flying and falling objects are present.

Confined Spaces

All job sites with potential confined spaces must have a Confined Space trained Competent Person on site to recognize confined space and implement required assessments and permits.

An excavation can be a confined space. A structure within an excavation (i.e., welding inside a pipeline or vault) often is also a confined space and workers must follow the confined space standards. To be cited for a trench or excavation as a confined space the excavation would have to meet the confined space definition in the standard and have hazards not addressed by the excavation standard.

The definition of a confined space is:

- (a) Large enough and arranged so an employee could fully enter the space and work.
- (b) Has limited or restricted entry or exit. Examples of spaces with limited or restricted entry are tanks, vaults, excavations, and pits.
- (c) Not primarily designed for continuous human occupancy.

EXCAVATION AND TRENCHING COMPETENT PERSON

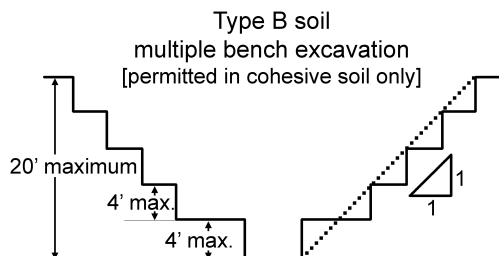
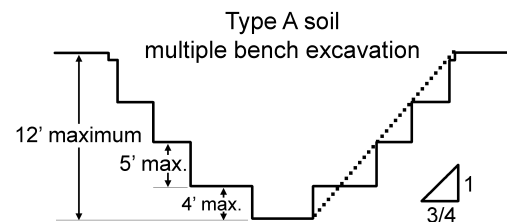
Protective Systems

Trenches that are deeper than 20' must have protective system that is designed by a registered professional engineer.

There are four types of Protection Systems that may be used in trenches greater than 4' (5' OSHA). The Competent Person supervising work must choose a protective system based on the work being performed.

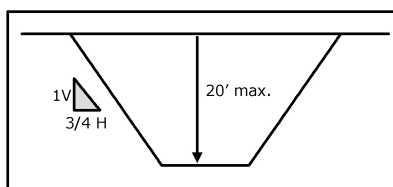
Benching: A method of protecting workers from cave-ins by excavating the sides of an excavation to form one or a series of horizontal levels or steps, usually with vertical or near-vertical surfaces between levels. *Benching cannot be done in Type C soil.*

BENCHING EXAMPLES:

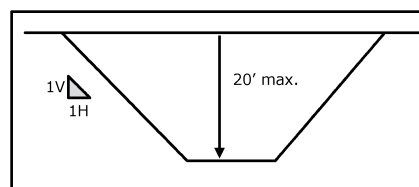


SLOPING EXAMPLES:

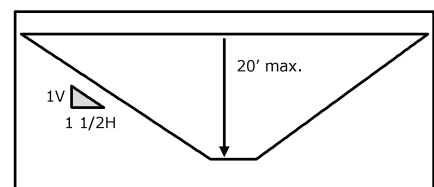
Excavations in type A soil



Excavations in type B soil



Excavations in type C soil



Sloping: The walls of the trench are sloped back during excavation to at least 3/4:1 for Class A, 1:1 for Class B, or 1 1/2:1 for Class C.

The determination of the angle of slope and design of the supporting system shall be based on careful evaluation by a "Competent Person" of pertinent factors, such as:

- Depth and/or cut/soils classification.
- Possible variation in water content of the material while excavation is open.
- Anticipated changes in materials from exposure to air, sun, water, or freezing.
- Loading imposed by structures, equipment, overlaying or stored material.
- Vibration from equipment, blasting, traffic, or other sources.

Surface encumbrances

You must remove or support surface encumbrances that are located so as to create a hazard to employees, as necessary, to safeguard employees.

Where the stability of adjoining buildings, walls, or other structures is endangered by excavation operations, you must provide support systems such as shoring, bracing, or underpinning to ensure the stability of such structures for the protection of employees.

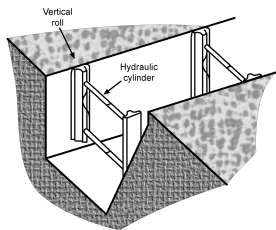
EXCAVATION AND TRENCHING COMPETENT PERSON

Shoring: Vertical walls of the trench are shored with lumber to prevent a cave in. Shoring must be designed by a Competent Person using established shoring charts based on soil type and depth in WAC 296-155-66405 Appendix C.

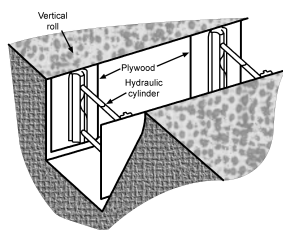
Other shoring systems, such as vertical aluminum shoring may be used following the shoring charts in WAC 296-155-66407 Appendix D.

Note: Poor quality lumber will often fail in a trench collapse. It is critical that the Competent Person uses appropriate material for shoring.

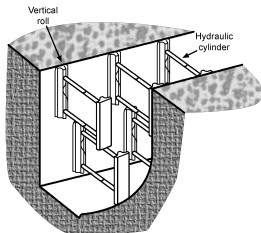
SHORING EXAMPLES:



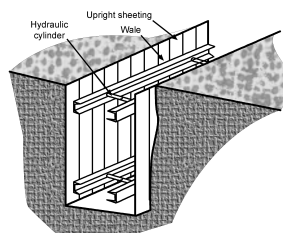
Vertical aluminum hydraulic shoring (spot bracing)



Vertical aluminum hydraulic shoring (with plywood)



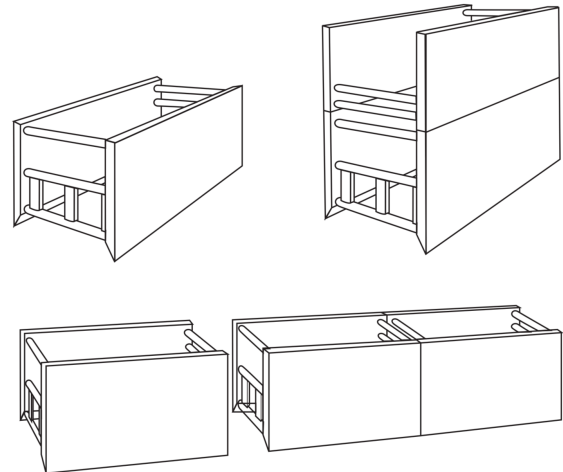
Vertical aluminum hydraulic shoring (stacked)



Aluminum hydraulic shoring water system (typical)

Shielding: Trench boxes designed for the soil type and depth of the trench are placed in the trench where the workers WILL be working. No worker is to be in the trench while the trench box is being placed by heavy equipment. Trench boxes need to be installed per the manufacturer's instructions. Trench boxes must have a ladder for egress when workers are working within the box. Trench boxes must be set at least level with the ground and no more than 2' off the bottom of the trench.

SHIELDING EXAMPLES:



Tabulated Data

All shoring and shielding systems must be designed using a registered professional engineer or tabulated data.

Tabulated data takes the soil type and depth of the trench and then lists materials for wales, cross bracing, and uprights in shoring systems. The tabulated data for shoring can be found in the WAC 296-155 part N tables and 1926 Subpart P tables.

Shield tabulated data is supplied by the manufacturer and will show allowable usage of each size of shield based on the depth and soil type. Some shields may be stacked for use in the field.

3L SERIES TRENCH SHIELD SIDEWALLS: sets of two									
MODEL	SIZE		PIPE CLEARANCE	WEIGHT (LBS)	SHIELD CAP. (PSF)(LBS)	MAXIMUM DEPTH PER SOIL TYPE (FT.) ¹			
	H	L				A	B	C-60	C-80
3L46	4'	6'	24"	652	1142	48	27	21	16
3L48	4'	8'	24"	815	827	35	20	16	12
3L410	4'	10'	24"	979	523	23	14	11	9
3L412	4'	12'	24"	1142	360	16	10	8	6
3L66	6'	6'	24"	933	1244	53	31	24	19
3L68	6'	8'	24"	1130	742	33	19	15	12
3L610	6'	10'	24"	1360	469	22	13	11	9
3L612	6'	12'	24"	1595	323	16	10	8	7
3L86	8'	6'	24"	1148	1125	49	29	23	18
3L88	8'	8'	24"	1451	699	32	20	16	13
3L810	8'	10'	24"	1743	442	22	14	11	10
3L812	8'	12'	24"	2040	304	16	11	9	8

EXCAVATION AND TRENCHING COMPETENT PERSON

Rescue:

The employer must plan for and provide prompt, safe removal or rescue of employees in the event of an emergency.

A sufficient work plan description of 'prompt, safe removal or rescue of workers in the event of an emergency' can include coordination with available local emergency services (e.g., 911) but must account for your worksite's proximity to these local emergency services and any potential geographic, topographic, or other obstacles that could prevent immediate access to the worksite. In remote or otherwise inaccessible locations where local emergency services cannot feasibly respond to an emergency in a timely manner, your plan cannot rely upon local emergency services as a primary means of 'prompt and safe removal or rescue.'

All employees working in the trench must be trained on the Trench Work Plan.

Most trench rescue involves shoring around the victim and hand digging them out of the trench. Once the victim is freed, they may be placed in a harness or basket stretcher and lifted out of the trench by a piece of equipment.

A knowledge of tabulated data, equipment, and shoring methods is key to cave-in rescue training.

